ROHIT KALE

AI Engineer & Architect | Computer Vision & GenAI | AWS Certified Solutions Architect

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PROFILE

AI Engineer and Architect with 4.5+ years of experience designing and building production computer vision, LLM and agentic systems end to end, from data and model training to the edge, cloud and product architecture that runs them. I focus on measurable results and controlled cost: blind evaluation before any accuracy claim, idempotent pipelines, and guardrails on GPU spend. I work mainly in Python, PyTorch and FastAPI on AWS and GCP. MS in Artificial Intelligence.

TECHNICAL SKILLS

Architecture: event-driven & batch ML pipelines, microservices, edge-to-cloud ingestion, LLM gateways with provider fallback, RAG reference designs, API design, multi-tenant RBAC, cost & security guardrails

Computer Vision: YOLOv8 / YOLO11 / YOLO26, RF-DETR, SAM 2 / SAM 3, ByteTrack, DeepSORT, RTMPose, PARSeq OCR, Kalman / RTS smoothing, homography & multi-camera calibration, OpenCV, Detectron2

GenAI & Agents: OpenAI, Gemini (VLM), Claude, AWS Bedrock; LangChain, LangGraph, Google ADK, MCP; RAG (OpenSearch k-NN, Qdrant, sqlite-vec); OpenAI Realtime API, LiveKit voice agents; Vertex AI fine-tuning

ML & Evaluation: PyTorch, Ultralytics, scikit-learn, interpretable models, blind / held-out evaluation, dataset building & annotation tooling, reinforcement learning (DQN)

Backend: Python, FastAPI, asyncio, Celery + Redis, SQLAlchemy (async), Alembic, Pydantic, WebSocket, REST, Pytest

Cloud & MLOps: AWS (EC2 GPU / spot, S3, Lambda, Step Functions, ECS / ECR, OpenSearch, CloudWatch), GCP (Vertex AI, Cloud Run Jobs, Cloud Functions, Workflows, GCS), Docker, GitHub Actions, NVIDIA Jetson, Cloudflare R2 / Stream

Data & Frontend: PostgreSQL, Supabase (Auth, RLS), Redis, Firebase; React, Next.js, TypeScript, Vite, Tailwind, shadcn/ui, Flutter

PROFESSIONAL EXPERIENCE

AI/ML Engineer Cellstrat AI — 2025 – Present

UBALL: AI BASKETBALL ANALYTICS (LATEST)

- Designed and built a **4-camera game-understanding pipeline**: YOLO26 / RF-DETR detection, RTMPose, jersey OCR, ByteTrack, cross-camera identity fusion and a ball-possession state machine that emits shot, rebound and turnover events. Architecture runs from Jetson edge capture through S3 and AWS GPU batch jobs.
- Evaluated it blind against human ground truth: 88–94% shot detection and 89–98% make/miss on 3 of 4 held-out games. Retraining the far-angle model raised its detection from 68% to 93%.
- Built Gemini VLM play narration, a Vertex AI fine-tuning pipeline on GCP, and the FastAPI / Supabase / Celery product backend.

CELLASSIST: VOICE AI FOR HEALTHCARE

- Built a voice AI agent for patient triage that connects Exotel telephony WebSockets to the **OpenAI Realtime API** for speech-to-speech calls in multiple languages.
- Kept conversations natural with 200 ms audio buffering, noise suppression, interruption handling and 24 kHz PCM16 audio. Deployed on AWS Mumbai for the lowest caller latency.

NXCHAT: ENTERPRISE RAG CHAT

- Built an enterprise chat system with **RAG** on LangChain and AWS OpenSearch vector search, routing across AWS Bedrock and OpenAI models.
- Added dataset management, file filtering and a versioned FastAPI REST API, deployed on AWS EC2 / S3 with a Pytest suite.

REXAI: GENAI FILE CLASSIFICATION

- Built a scalable classification backend that combines lexical techniques with **LLM-based classification** to keep high-volume work accurate and cheap.
- Ran processing asynchronously on AWS Lambda + Step Functions with OpenSearch indexing for high-volume workloads.

CELLBOT: AI CHATBOT & REPORTS

- Built the FastAPI backend (Poetry, Pydantic, WeasyPrint) that generates PDF reports from conversations.
- Built the Next.js frontend with Clerk authentication, shadcn/ui, real-time markdown rendering and conversation history.

Software Engineer **Independent Consultant** Aug 2023 – 2024

- Built domain-specific ChatGPT chatbots and PayPal subscription billing (Supabase) for clients' FlutterFlow apps.
- Automated an end-to-end reading service with n8n + OpenAI: Razorpay payment, form extraction, generated documents and Gmail delivery, with retries and error handling.
- Containerised AI services with Docker and deployed them on AWS (Lambda, S3, CloudWatch) with GitHub Actions CI/CD.

Analyst **TheMathCompany, Bangalore** Feb 2022 – Aug 2023

- Built data pipelines for ML model training and evaluation for retail and BFSI clients, plus API-driven automation for client data workflows.
- Built dashboards that tracked system performance and data-quality metrics for stakeholders.

EDUCATION & CERTIFICATION

AWS Certified Solutions Architect – Associate, Amazon Web Services 2025

MS, Artificial Intelligence, Liverpool John Moores University 2021 – 2023

Post Graduate Diploma, Machine Learning & AI, IIIT Bangalore 2021 – 2022

B.Tech, Visvesvaraya National Institute of Technology (VNIT), Nagpur 2016 – 2020